Jeremeys, II

Day-2

"God's Unchanging Word for the Changing World"

Date:..... Time:..... Venue:.....

Interface & Cables

ⓐ Ethernet

↳ Ethernet is a collection of network protocols/standards.

ⓑ Network Protocol

ⓒ Bit → Represented by 0 or 1

ⓓ Bytes → 8 bit = 1 byte

Speed is measured in bits per second.

Data is measured in byte.

- 1 kilobit (kb) = 1000 bits
- 1 megabit (Mb) = 1000,000 bits
- 1 gigabit (Gb) = 1,000,000,000 bits
- 1 ~~Tera~~ Terabit (Tb) = 1 000,000,000,000 bits

Ethernet Standards

- Defined in the IEEE 8023

IEEE = Institute of Electrical & Electronics Engineers

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Ethernet Standards (Copper)

Speed	Common Name	IEEE Standard	Informal Name	Maximum Length
10 Mbps	Ethernet	802.3i	10BASE-T	100m
100 Mbps	Fast Ethernet	802.3u	100BASE-T	100m
1 Gbps	Gigabit Ethernet	802.3ab	1000BASE-T	100m
10 Gbps	10 Gig Ethernet	802.3an	10GBASE-T	100m

BASE = refers to baseband signaling

T = twisted pair

UTP Cables (Unshielded Twisted Pair)

→ Affected by electromagnetic interference

→ 4 pairs

10 BASE-T = 2 pairs (4 wires)

100 BASE-T

1000 BASE-T = 4 pairs (8 wires)

10 GBASE-T

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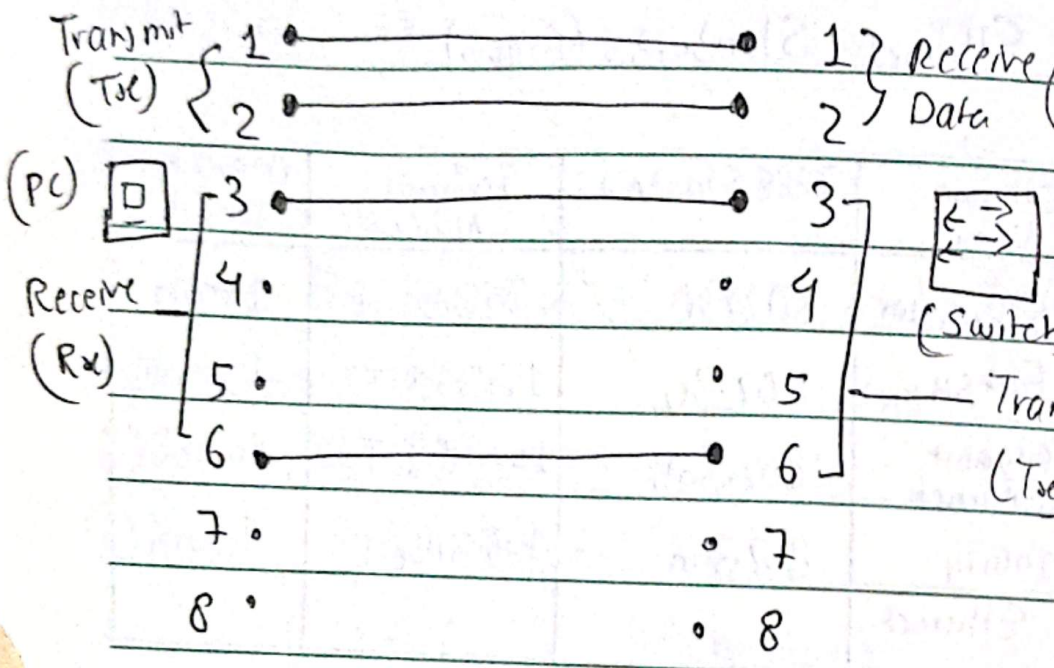
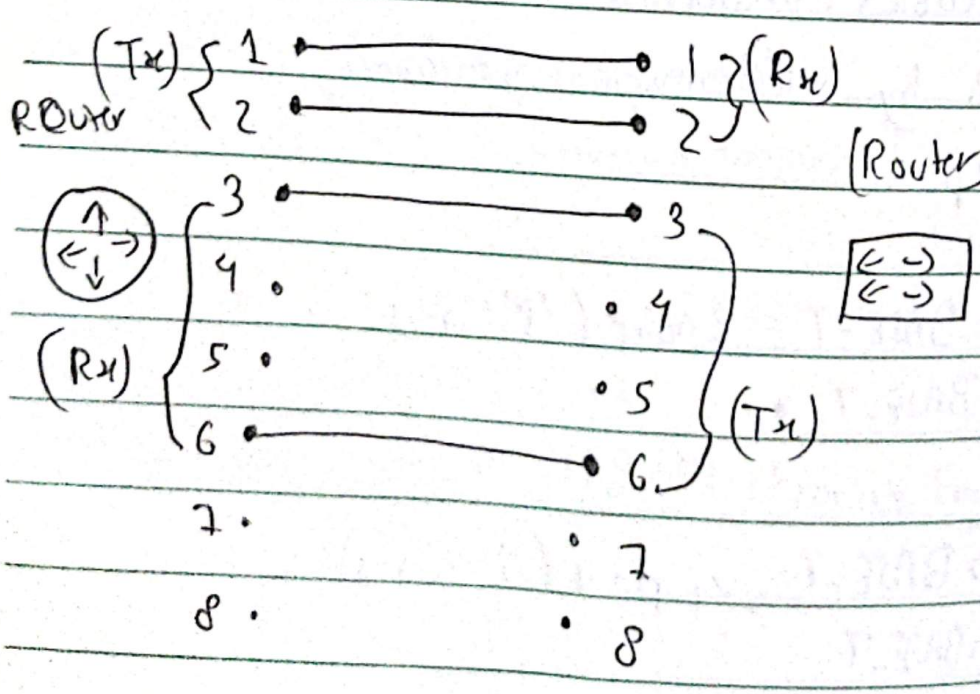


10 BASE-T, 100BASE-T
(Full-Duplex Transmission)



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So: PC transmits from 1 and 2
Receives from 3 & 6

Router transmit from ~~3 & 6~~ 1 & 2
& Receives from 3 & 6
~~Switch~~

Switch transmits from 3 & 6
Receives from 1 & 2.

Memory :- PC & Router Tx - Rx
Switch Rx - Tx

⊙ These kind of cable is called straight-through cable

For connecting same devices example Router to a Router or switch to a switch we use cross-over cable

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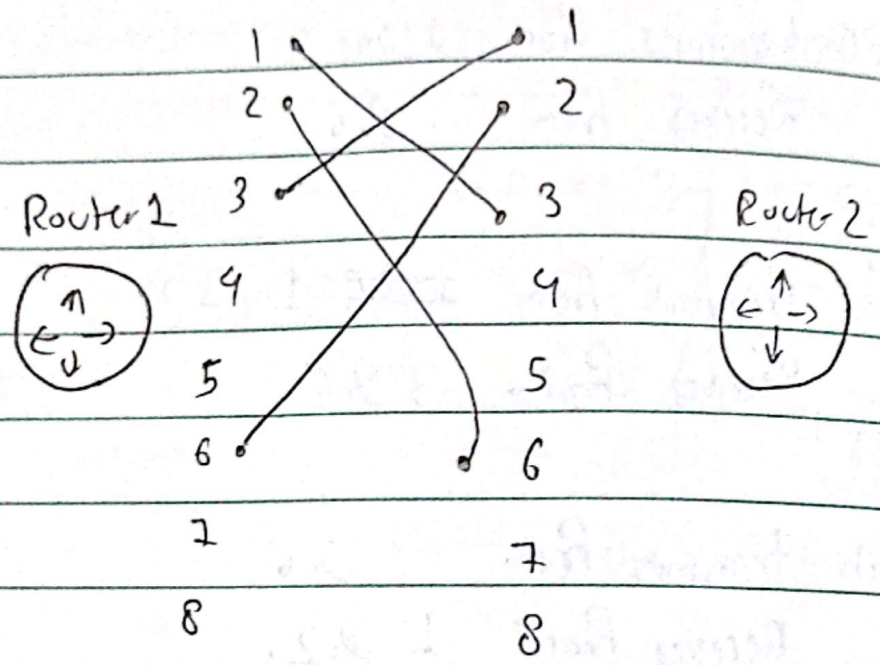
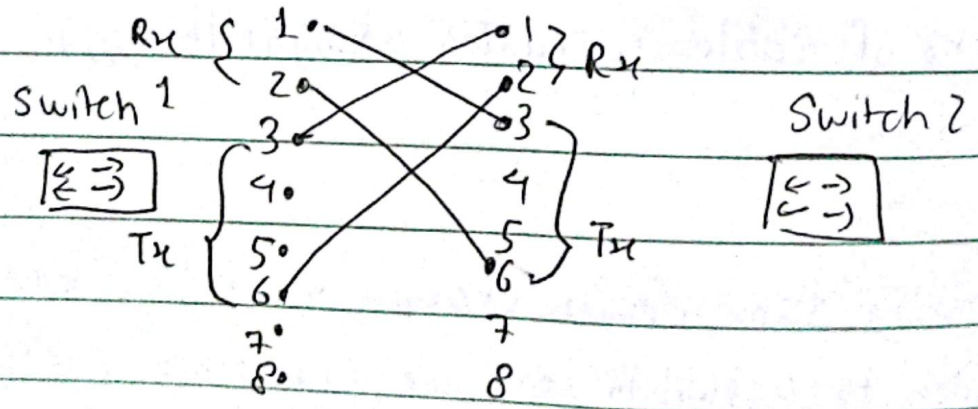


Router 1 transmits from 1 & 2
Router 2 Receives from 3 & 6 (vice versa)



Switch uses 3 & 6 to transmit and 1 & 2 to receive.
So, Switch 1 → 3 & 6 (Tx) to 1 & 2 (Rx)

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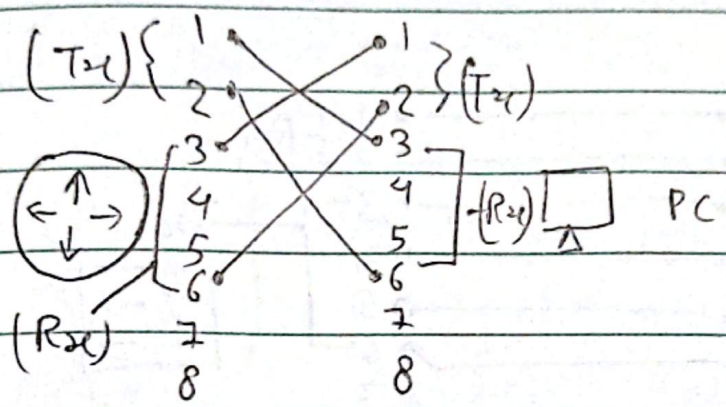

Both uses 1 & 2 for (Tx) and 3 & 6 for receive

So we use Cross over cable

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Device Type	Tx Pins	Rx Pins
Router	1 & 2	3 & 6
Firewall	1 & 2	3 & 6
PC	1 & 2	3 & 6
Switch	3 & 6	1 & 2

Auto MDI-X (Medium Dependent Interface Crossover)

↳ This allows modern device to communicate with same device with straight through cable.

It detects which pins is sending data & receiving data.

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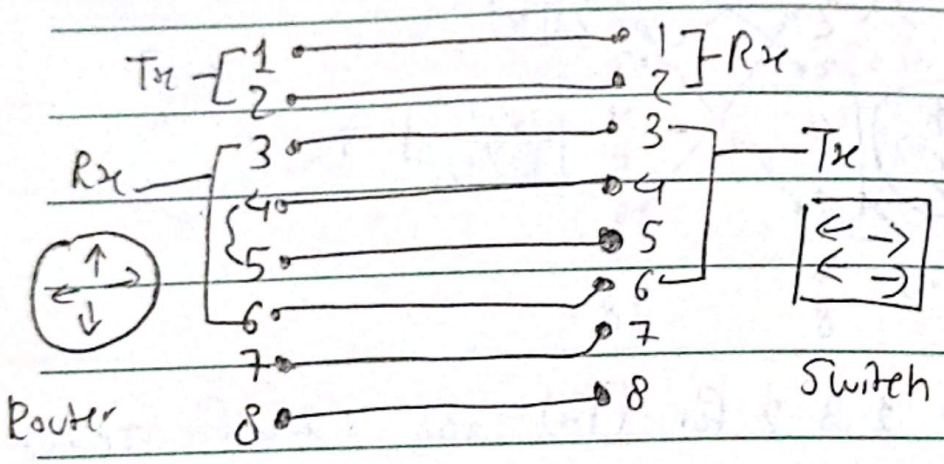
UTP Cable (1000BASE-T & 10GBASE-T)



All 8 pins are used in this Router & Switch

4 & 5 => Each pair is bidirectional

7 & 8 => "

1 & 2 => "

3 & 6 => "

Fiber-Optics Connections

SFP => Small Form-Factor Pluggable Transceiver

We connect fiber optic cable through SFP.

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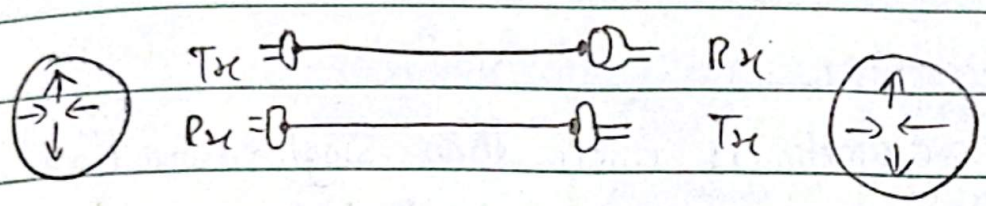
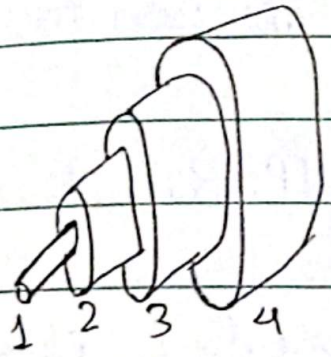


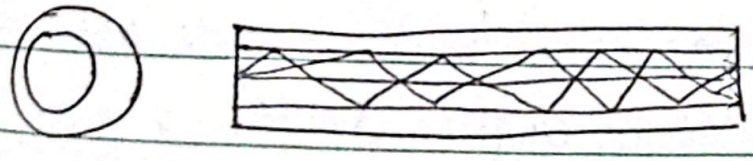


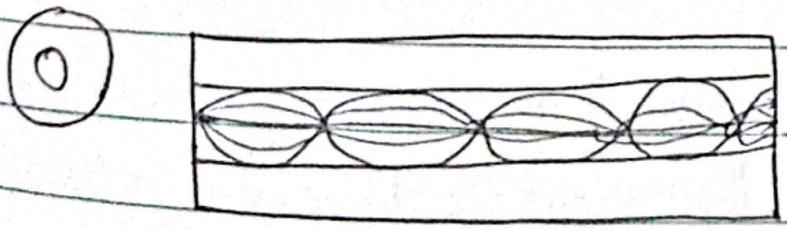
1- Fiberglass core itself
(light is transmitted here)
2- Cladding that reflects light
3- a protective buffer
(Protects fiber glass from breaking)
4- Outer Jacket of the cable

Types of fiber optics

① Single-mode ② Multimode

Multimode fiber. Cables.





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Characteristics

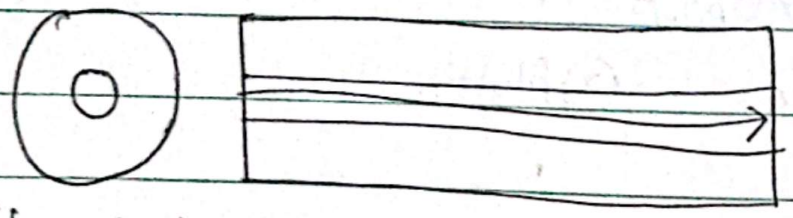
① Core-diameter is wider than single-mode fiber.

② Allows multiple angles(modes) of light waves to enter the fiberglass core

③ Allows longer cables than UTP. But shorter cables than single-mode fiber.

④ Cheaper than single-mode fiber (due to cheaper LED based SFP Transmitter).

• Single-Mode fiber



Characteristics

① Core diameter is narrower than multimode fiber.

② Light Enters at a single angle(mode) from a laser-based transmitter.

③ Allows longer cables that UTP and multimode fiber.

④ More expensive than multimode fiber (due to more expensive laser based SFP transmitters)

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Fiber Optics Cable Standards

Informal Names	IEEE Standards	Speed	Cable Type	Maximum Length
1000 BASE-LX	802.3z	1Gbps	Multimode or Singlemode	550m (MM) 5km (SM)
10GBASE-SR	802.3ae	10Gbps	Multimode	400m
10GBASE-LR	802.3ae	10Gbps	Singlemode	10km
10GBASE-ER	802.3ae	10Gbps	Singlemode	30km

± Differences between UTP & Fiber Optics

UTP	Fiber Optics
① Lower Cost	① Expensive
② Shorter maximum distance than fiber optics.	② Longer Distances.
③ Vulnerable to EMI	③ No Vulnerability to EMI
④ RJ45 ports used with UTP are cheaper than SFP ports	④ SFP Ports are more expensive. Single mode is more expensive.
⑤ Emit (leaks) a faint signal outside of a cable, which can be copied (= security risk)	⑤ Does not emit any signal outside of the cable (no security risk.)

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